Day 068 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Write and Graph Inequalities

Standards: 6.EE.B.8

Learning target: I can write and graph a constraint using an inequality.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 25

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01 / Learn

Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions.

Worked example 1

Let x be any real number. Write and graph: x is more than 48. Label the endpoint and shade the solution ray.

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. Graph a strict bound with an open circle; use a closed circle for an included endpoint.

3. “more than” means >

4. exclude the endpoint.

5. State the result in the form requested: x > 48; open circle at 48, shade right.

Answer: x > 48; open circle at 48, shade right

Worked example 2

A club needs at least 35 members. Select all true statements for its membership count n.
A. 35 is an allowed membership count.
B. 34 is an allowed membership count.
C. n > 35
D. n ≥ 35

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. At least includes the boundary, so n ≥ 35.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - 35 is an allowed membership count.; n ≥ 35

Day 068 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Let x be any real number. Write and graph: x is at most 49. Label the endpoint and shade the solution ray.

02 / Constructed response

Let x be any real number. Write and graph: x is at least 42. Label the endpoint and shade the solution ray.

03 / Constructed response

Let x be any real number. Write and graph: x is at least 35. Label the endpoint and shade the solution ray.

04 / Constructed response

Let x be any real number. Write and graph: x is at most 16. Label the endpoint and shade the solution ray.

05 / Constructed response

Let x be any real number. Write and graph: x is more than 8. Label the endpoint and shade the solution ray.

06 / Constructed response

Let x be any real number. Write and graph: x is more than 25. Label the endpoint and shade the solution ray.

07 / Constructed response

Let x be any real number. Write and graph: x is at least 24. Give one included value and explain whether the endpoint is included.

08 / Multiple select

A club needs at least 10 members. Select all true statements for its membership count n.
A. n > 10
B. 10 is an allowed membership count.
C. 9 is an allowed membership count.
D. n ≥ 10

09 / Multiple select

A club needs at least 8 members. Select all true statements for its membership count n.
A. 7 is an allowed membership count.
B. n ≥ 8
C. 8 is an allowed membership count.
D. n > 8

10 / Constructed response

A museum group must have at least 25 participants. Let p be the participant count. Write an inequality and graph its real-number solution set. State which solutions make sense as participant counts.

Day 068 | Unit 05 | Answer key

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1. x ≤ 49; closed circle at 49, shade left

“at most” means ≤; include the endpoint.

6.EE.B.8 | Constructed response

2. x ≥ 42; closed circle at 42, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

3. x ≥ 35; closed circle at 35, shade right

“at least” means ≥; include the endpoint.

6.EE.B.8 | Constructed response

4. x ≤ 16; closed circle at 16, shade left

“at most” means ≤; include the endpoint.

6.EE.B.8 | Constructed response

5. x > 8; open circle at 8, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

6. x > 25; open circle at 25, shade right

“more than” means >; exclude the endpoint.

6.EE.B.8 | Constructed response

7. x ≥ 24; closed circle at 24, shade right

“at least” means ≥; include the endpoint. Example solution: 25.

6.EE.B.8 | Constructed response

8. B, D - 10 is an allowed membership count.; n ≥ 10

At least includes the boundary, so n ≥ 10. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

9. B, C - n ≥ 8; 8 is an allowed membership count.

At least includes the boundary, so n ≥ 8. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

10. p ≥ 25; closed circle at 25, shade right; participant counts are whole numbers 25, 26, …

“at least” means ≥. The real-number ray has infinitely many solutions, but participant counts must be whole numbers.

6.EE.B.8 | Constructed response